

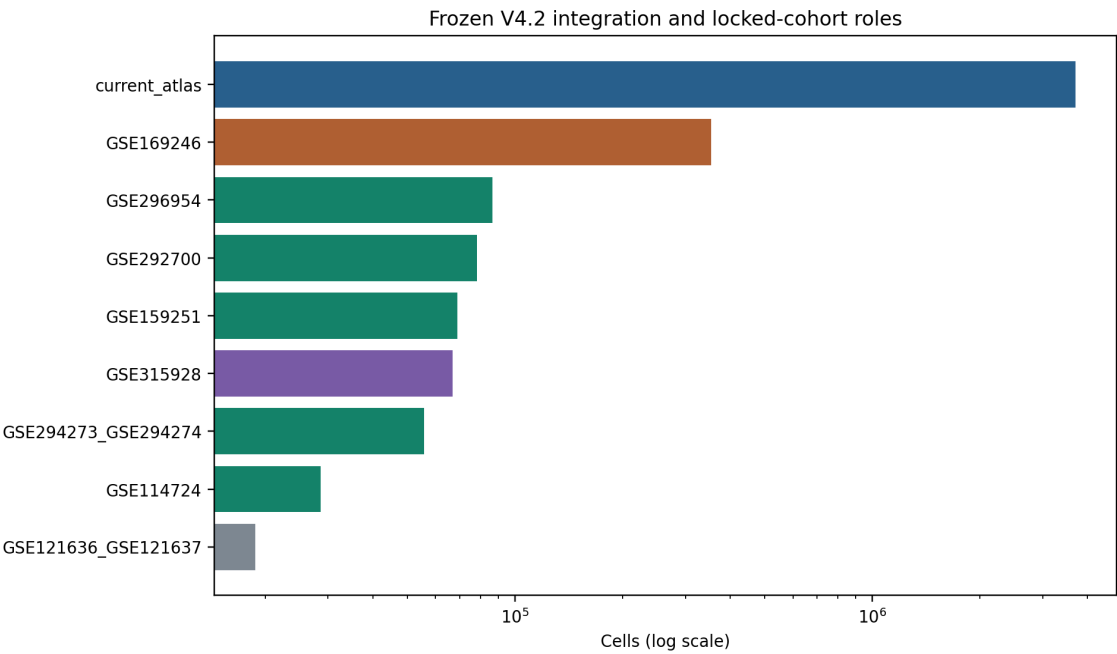
gdTAI V4.2 integration preflight

gdTAI V4.2 modeling-integration preflight

Decision

PASS_REVIEW_REQUIRED. The proposed sidecar integration is technically feasible and the whole-dataset role split is internally consistent. This report authorizes supervision review only. It does not authorize integration, scVI fitting, clustering, pseudo-labeling, classifier fitting, threshold selection, promotion, or atlas inference.

- Development integration: **4,023,462 cells** from the current atlas plus five new development cohorts.
- Locked evaluation: **439,979 cells** across three whole cohorts excluded before HVG selection and scVI fitting.
- Common development genes: **14,265**; eligible after TCR/mitochondrial/ribosomal/immunoglobulin exclusion: **13,975**.
- GSE169246 contains all 50 Stage-1 genes but only **145/197 (73.6%)** full classifier genes. It is a valid Stage-1 NK-passage challenge but only a reduced-feature sensitivity analysis for final-cascade FPR.
- The extension cohorts contain no expression-independent gdT-positive truth. They strengthen specificity training/evaluation but cannot establish gdT recall, F1, or superiority alone.



Cohort roles

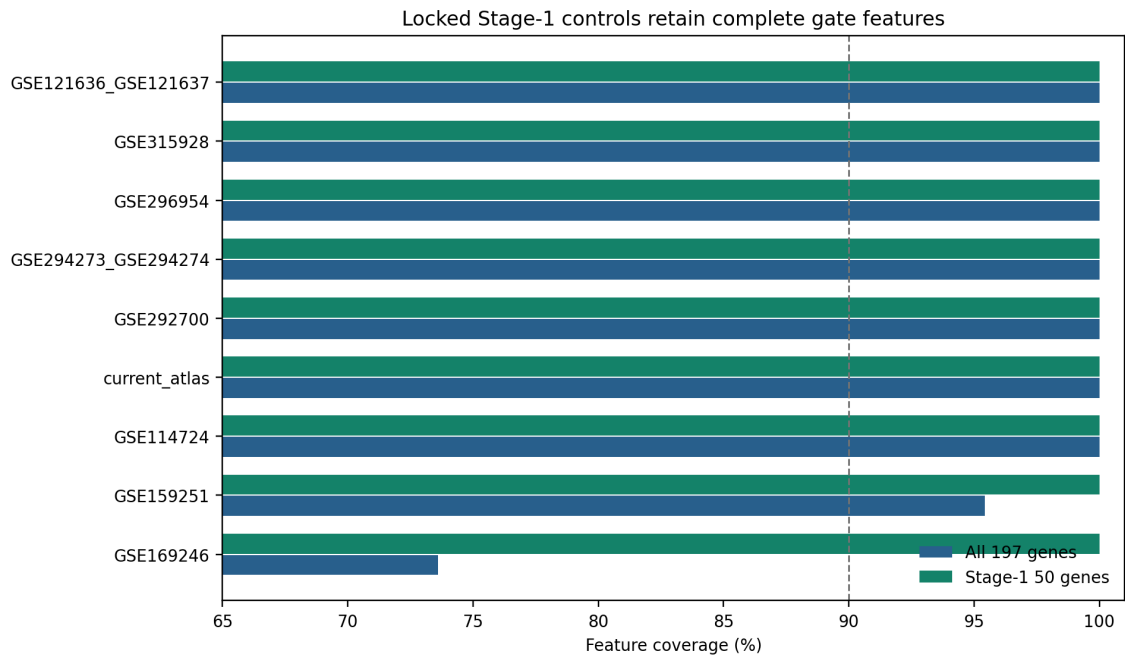
Frozen cohort roles

cohort_id	role	include_in_integration_fit	allow_pseudolabel_training	allow_model_tuning	allow_locked_evaluation	expected_cells
current_atlas	development_reference	True	False	True	False	3705306
GSE114724	development_cluster_expansion	True	True	True	False	28652
GSE159251	development_cluster_expansion	True	True	True	False	68898
GSE292700	development_cluster_expansion	True	True	True	False	78254
GSE294273_GSE294274	development_cluster_expansion	True	True	True	False	55744
GSE296954	development_cluster_expansion	True	True	True	False	86608
GSE169246	locked_author_nk_stage1_specificity	False	False	False	True	354373
GSE315928	locked_paired_abt_specificity	False	False	False	True	66813
GSE121636_GSE121637	locked_mixed_t_nk_stress	False	False	False	True	18793

All locked cohorts are excluded from HVG selection, scVI training, clustering, cluster interpretation, pseudo-label construction, feature selection, calibration, thresholding, and model-family selection. These cohorts have already been inspected by earlier frozen-model screens and are not prospective or fully independent; they are locked specifically for V4.2 comparison.

Matrix and feature compatibility

cohort_id	n_cells	n_genes	matrix_key	matrix_encoding	matrix_dtype	sample_integer_like	model_feature_coverage	stage1_feature_coverage
current_atlas	3705306	27413	X	csr_matrix	float32	True	100.0%	100.0%
GSE114724	28652	33694	X	csr_matrix	int64	True	100.0%	100.0%
GSE159251	68898	22013	X	csr_matrix	int64	True	95.4%	100.0%
GSE292700	78254	38606	X	csr_matrix	int64	True	100.0%	100.0%
GSE294273_GSE294274	55744	33540	X	csr_matrix	int64	True	100.0%	100.0%
GSE296954	86608	36601	X	csr_matrix	float32	True	100.0%	100.0%
GSE169246	354373	12989	layers/counts	csr_matrix	int64	True	73.6%	100.0%
GSE315928	66813	36601	X	csr_matrix	float32	True	100.0%	100.0%
GSE121636_GSE121637	18793	33694	X	csr_matrix	int64	True	100.0%	100.0%



Feature coverage

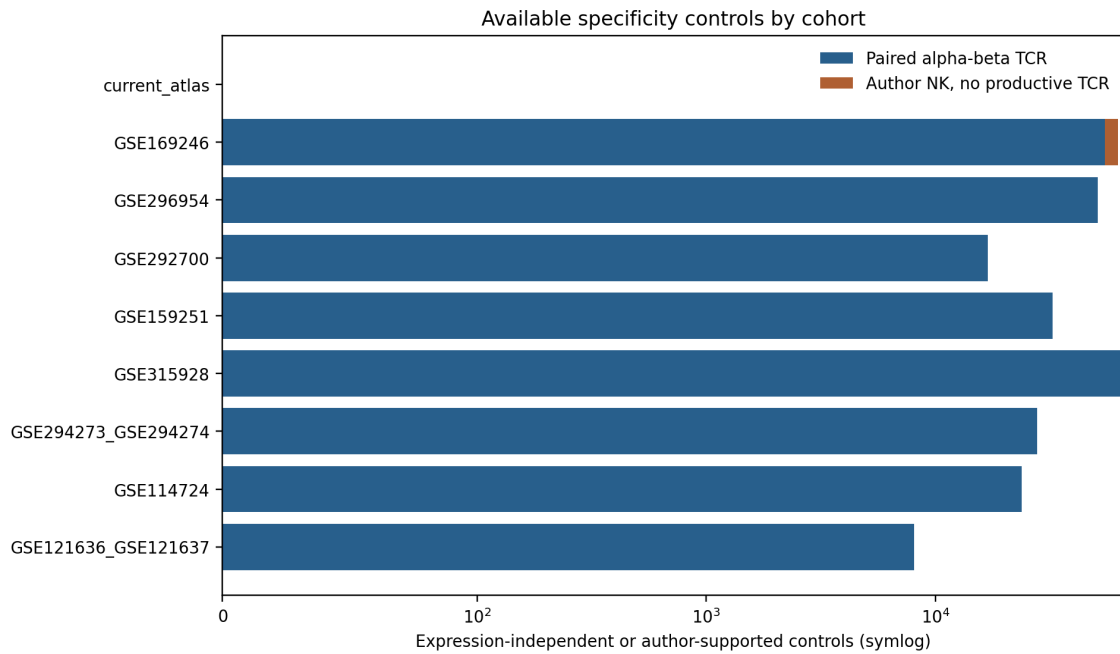
The development intersection supports 4,000 batch-aware HVGs without forcing classifier genes into the latent representation. All TCR genes are excluded from HVG selection so that TCR-family expression cannot directly define cluster-derived NK pseudo-labels. The classifier itself remains a log1p(CP10K) individual-gene model and never consumes scVI coordinates or Leiden labels.

Available controls

cohort_id	n_cells	n_paired_ab_tcr	n_paired_gd_tcr	n_author_nk_no_productive_tcr
current_atlas	3705306	0	0	0
GSE114724	28652	23789	0	0
GSE159251	68898	32420	0	0
GSE292700	78254	16903	0	0
GSE294273_GSE294274	55744	27838	0	0
GSE296954	86608	51063	0	0
GSE169246	354373	54925	0	7770
GSE315928	66813	66813	0	0
GSE121636_GSE121637	18793	8094	0	0

Primary weight-1 NK anchors mapped to the current atlas:

source_gse_id	n_anchor_cells	n_matched_current_atlas
GSE125527	3547	3547
GSE228597	17507	17507



Specificity controls

GSE169246 contributes 7,770 author NK cells without productive TCR and 54,925 productive paired alpha-beta T cells. GSE315928 contributes 66,813 paired alpha-beta T cells with complete 197-gene coverage. GSE121636/GSE121637 remains a mixed stress cohort because it lacks an independent author NK label in the harmonized object.

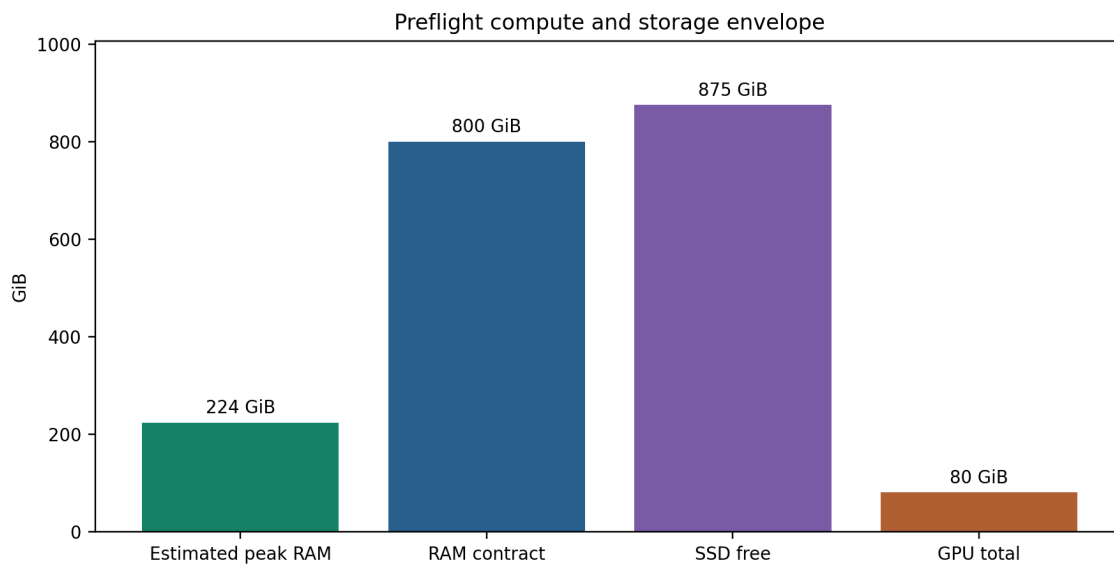
Precommitted integration and clustering

1. Build a new sidecar object under the mirrored SSD root; never mutate a source or canonical milestone H5AD.
2. Use raw counts, 4,000 batch-aware common HVGs, the existing library/sample/source batch hierarchy, scVI 128 x 2, 30 latent dimensions, negative-binomial likelihood, 20 epochs maximum, batch size 8,192, GPU only, and early stopping.
3. Run nine global RAPIDS Leiden partitions: resolutions 0.4, 0.8, 1.2 crossed with seeds 11, 29, 47.
4. Recluster the T/NK boundary in nine additional partitions: resolutions 0.3, 0.6, 1.0 crossed with the same seeds.
5. A cluster-derived development NK negative requires $\geq 80\%$ run agreement, $\geq 95\%$ independently labeled NK-anchor purity, $\leq 2\%$ productive-TCR contamination, representation in ≥ 3 development sources, and $\leq 70\%$ contribution from any one source.
6. No threshold on CD3, TRDC, TRDV, NKG7, GNLY, KLRD1, TYROBP, FCER1G, or FCGR3A may define truth. Pseudo-labels receive weight 0.5, are source-balanced and capped, and cannot determine validation guardrails.

Primary paired-gdT inputs remain outside integration and pseudo-label construction. They retain weight-1 positive roles during classifier fitting. Their later latent-query proximity to NK clusters is diagnostic only and cannot tune the pseudo-label rules.

Resource feasibility

development_cells	input_full_csr_gib	estimated_common_csr_gib	estimated_hvg_csr_gib	latent_gib	knn_graph_gib	conservative_peak_ram_gib	ssd_free_gib	gpu_name	gpu_total_gib
4023462	66.6	34.7	9.7	0.45	1.35	223.6	875.1	NVIDIA A100 80GB PCIe	80



Resources

The estimate is intentionally conservative and remains below the active 800-GiB exception. The A100 is available with more than the frozen 75-GiB minimum, and SSD free space exceeds the 300-GiB floor.

QC checks

check	status	detail
sha256::current_atlas	PASS	9060920d6fe19bdbbc1845d79a571ce545bfe731642a83550e6f784fe3f7d2c3
expected_cells::current_atlas	PASS	3705306
sparse_matrix::current_atlas	PASS	csr_matrix
raw_count_sample::current_atlas	PASS	n=90000; range=1.0..1866.0
unique_axes::current_atlas	PASS	cell_duplicates=0; gene_duplicates=0
sha256::GSE114724	PASS	394425c7937432ac7fce4b904b35b46fa7519fc5107e4942aff27cf7d6e7a808
expected_cells::GSE114724	PASS	28652
sparse_matrix::GSE114724	PASS	csr_matrix
raw_count_sample::GSE114724	PASS	n=90000; range=1.0..218.0
unique_axes::GSE114724	PASS	cell_duplicates=0; gene_duplicates=0
sha256::GSE159251	PASS	abe6502b3a15a3ba458fc47b46e86277e8b9764665722616205201f8452ed569
expected_cells::GSE159251	PASS	68898
sparse_matrix::GSE159251	PASS	csr_matrix
raw_count_sample::GSE159251	PASS	n=90000; range=1.0..196.0
unique_axes::GSE159251	PASS	cell_duplicates=0; gene_duplicates=0
sha256::GSE292700	PASS	78c9ada980a21592ba72a56c1beea12dcd0d256a630b2294be68b711d977dc8a
expected_cells::GSE292700	PASS	78254
sparse_matrix::GSE292700	PASS	csr_matrix
raw_count_sample::GSE292700	PASS	n=90000; range=1.0..459.0
unique_axes::GSE292700	PASS	cell_duplicates=0; gene_duplicates=0
sha256::GSE294273_GSE294274	PASS	6a20d61544c7cd33d33ceec756837308e7a88d992919e6496a29822eda9215cf
expected_cells::GSE294273_GSE294274	PASS	55744
sparse_matrix::GSE294273_GSE294274	PASS	csr_matrix
raw_count_sample::GSE294273_GSE294274	PASS	n=90000; range=1.0..699.0
unique_axes::GSE294273_GSE294274	PASS	cell_duplicates=0; gene_duplicates=0
sha256::GSE296954	PASS	f3484482e35b49d3c3f192ad1e6570d2c066f3cf68db99b5bafa9312dc440b0
expected_cells::GSE296954	PASS	86608
sparse_matrix::GSE296954	PASS	csr_matrix
raw_count_sample::GSE296954	PASS	n=90000; range=1.0..523.0
unique_axes::GSE296954	PASS	cell_duplicates=0; gene_duplicates=0
sha256::GSE169246	PASS	f251382c62109a486500a2ac472e2e5608d368625891db3de512956bd5ce6e6c
expected_cells::GSE169246	PASS	354373
sparse_matrix::GSE169246	PASS	csr_matrix
raw_count_sample::GSE169246	PASS	n=90000; range=1.0..179.0
unique_axes::GSE169246	PASS	cell_duplicates=0; gene_duplicates=0
sha256::GSE315928	PASS	2fe674761db67aa337c4d80932d69d4bfb47e44d2a16c22cbf5ecd819c8d70d
expected_cells::GSE315928	PASS	66813

check	status	detail
sparse_matrix::GSE315928	PASS	csr_matrix
raw_count_sample::GSE315928	PASS	n=90000; range=1.0..257.0
unique_axes::GSE315928	PASS	cell_duplicates=0; gene_duplicates=0
sha256::GSE121636_GSE121637	PASS	b7182274cab834f10e0eb8de240c7d770c5f8890be0fe862d41712f7476e463d
expected_cells::GSE121636_GSE121637	PASS	18793
sparse_matrix::GSE121636_GSE121637	PASS	csr_matrix
raw_count_sample::GSE121636_GSE121637	PASS	n=90000; range=1.0..789.0
unique_axes::GSE121636_GSE121637	PASS	cell_duplicates=0; gene_duplicates=0
unique_cohort_roles	PASS	duplicates=0
locked_cohorts_excluded_from_development	PASS	leaking_cohorts=0
development_source_count	PASS	6 input objects
primary_nk_anchor_count	PASS	21054 anchors
primary_nk_anchor_source_count	PASS	2 sources
primary_nk_anchors_map_to_current_atlas	PASS	21054/21054 matched
common_gene_universe	PASS	13975 eligible common genes
development_model_feature_coverage	PASS	minimum=0.954
locked_stage1_feature_coverage	PASS	minimum=1.000
required_metadata_columns	PASS	missing=0
ram_feasibility	PASS	estimated 223.6 GiB < 800 GiB
ssd_feasibility	PASS	875.1 GiB free
gpu_feasibility	PASS	NVIDIA A100 80GB PCIe; total=80.0 GiB
all_input_h5ads_unchanged	PASS	9/9 unchanged

Missing required metadata columns: **0**. All nine input H5AD size and nanosecond modification-time pairs were checked before and after the audit.

Supervision gate

The next executable milestone is implementation QC for a fail-closed sidecar integration runner. It requires explicit approval. Even after integration approval, V4.2 classifier fitting requires a separate reviewed QC gate and approval record.

Outputs

- Full tables: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_2_integration_preflight/`
- Machine summary: `Integrated_dataset/logs/gdT_prediction/gdtai_v4_2_integration_preflight/summary.json`
- HTML: `gdT_prediction/gdtai_v4_2_integration_preflight/index.html`
- PDF: `gdT_prediction/gdtai_v4_2_integration_preflight/gdtai_v4_2_integration_preflight_report.pdf`